

OSSP (Operating Systems And Systems Programming) - 25CS2104E

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SEC-8

PRACTICAL-1:

Aim: Develop a C program that demonstrates how a Linux operating system executes a command entered by a user that 1. Accept a Linux command as input. 2. Create a child process using fork(). 3. Execute the command in the child process using an appropriate exec() system call. 4. Allow the parent process to wait for the child using wait (). 5. Display the Process ID (PID) of both parent and child processes.

Using Linux terminal commands (uname, lscpu, lsblk, ps, top), investigate the relationship between hardware resources and operating system services. Prepare a report explaining how the OS abstracts CPU, memory, storage, and I/O devices.

```
Windows PowerShell
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PS C:\Users\Admin> wsl
admin2007@DESKTOP-MPC8TJM:/mnt/c/Users/Admin$ cd /mnt/c/OS_C_Files/Practical-1
admin2007@DESKTOP-MPC8TJM:/mnt/c/OS_C_Files/Practical-1$ ls
Case01  Readme.txt  a.out  build  case01.C  run.bat
admin2007@DESKTOP-MPC8TJM:/mnt/c/OS_C_Files/Practical-1$ gcc -x c case01.C -o case01
admin2007@DESKTOP-MPC8TJM:/mnt/c/OS_C_Files/Practical-1$ ./case01
Enter command: ls
Parent PID: 427
Child PID: 428
Readme.txt a.out build case01 case01.C run.bat
Child finished. Parent PID: 427
admin2007@DESKTOP-MPC8TJM:/mnt/c/OS_C_Files/Practical-1$ ./case01
Enter command: pwd
Parent PID: 429
Child PID: 430
/mnt/c/OS_C_Files/Practical-1
Child finished. Parent PID: 429
admin2007@DESKTOP-MPC8TJM:/mnt/c/OS_C_Files/Practical-1$ |
```

```
admin2007@DESKTOP-MPC8TJM:/mnt/c/OS_C_Files/Practical-1$ ./case01
Enter command: uname
Parent PID: 391
Child PID: 392
Linux
Child finished. Parent PID: 391
```

```
0[          0.0%] 4[          0.0%]
1[          0.0%] 5[          0.4%]
2[          0.0%] 6[          0.4%]
3[          0.0%] 7[          0.0%]
Mem[|||||] 366M/7.46G Tasks: 24, 14 thr, 0 kthr; 2 running
Swp[          0K/2.00G Load average: 0.10 0.05 0.01
Uptime: 00:02:14

Main 1/0
PID USER      PRI  NI  VIRT   RES   SHR  S  CPU% MEM%   TIME+  Command
1 root         20   0 21648 12392 9320 S   0.0  0.2  0:01.06 /sbin/init
2 root         20   0 3120  1920  1920 S   0.0  0.0  0:00.00 /init
6 root         20   0 3120  1792  1792 S   0.0  0.0  0:00.00 plan9 --control-socket 7 --log-level 4 --server-fd 8 --pipe-fd 10
7 root         20   0 3120  1792  1792 S   0.0  0.0  0:00.00 plan9 --control-socket 7 --log-level 4 --server-fd 8 --pipe-fd 10
8 root         20   0 3120  1920  1920 S   0.0  0.0  0:00.00 /init
42 root        19  -1 50436 15100 14204 S   0.0  0.2  0:00.29 /usr/lib/systemd/systemd-journald
93 root         20   0 25284 6400  4992 S   0.0  0.1  0:00.16 /usr/lib/systemd/systemd-udev
147 systemd-re 20   0 21460 12800 10624 S   0.0  0.2  0:00.11 /usr/lib/systemd/systemd-resolved
154 systemd-ti 20   0 91028 7680  6784 S   0.0  0.1  0:00.06 /usr/lib/systemd/systemd-timesyncd
158 systemd-ti 20   0 91028 7680  6784 S   0.0  0.1  0:00.00 /usr/lib/systemd/systemd-timesyncd
163 root         20   0 4236  2560  2432 S   0.0  0.0  0:00.00 /usr/sbin/cron -f -P
164 messagebus 20   0 9628  4864  4480 S   0.0  0.1  0:00.05 @dbus-daemon --system --address=systemd: --nofork --nopidfile --sy
171 root         20   0 17972  8320  7552 S   0.0  0.1  0:00.08 /usr/lib/systemd/systemd-logind
173 root         20   0 1715M 12672 10880 S   0.0  0.2  0:00.08 /usr/libexec/wsl-pro-service
176 syslog      20   0 217M  5248  4480 S   0.0  0.1  0:00.06 /usr/sbin/rsyslogd -n -iNONE
182 root         20   0 3160  1792  1792 S   0.0  0.0  0:00.00 /sbin/agetty -o -p -- \u --noclear --keep-baud - 115200,38400,9600
189 root         20   0 3116  1920  1792 S   0.0  0.0  0:00.01 /sbin/agetty -o -p -- \u --noclear - linux
197 root         20   0 104M 22264 13056 S   0.0  0.3  0:00.16 /usr/bin/python3 /usr/share/unattended-upgrades/unattended-upgrade
198 root         20   0 1715M 12672 10880 S   0.0  0.2  0:00.00 /usr/libexec/wsl-pro-service
199 syslog      20   0 217M  5248  4480 S   0.0  0.1  0:00.00 /usr/sbin/rsyslogd -n -iNONE
200 syslog      20   0 217M  5248  4480 S   0.0  0.1  0:00.00 /usr/sbin/rsyslogd -n -iNONE
201 syslog      20   0 217M  5248  4480 S   0.0  0.1  0:00.00 /usr/sbin/rsyslogd -n -iNONE
202 root         20   0 1715M 12672 10880 S   0.0  0.2  0:00.00 /usr/libexec/wsl-pro-service
203 root         20   0 1715M 12672 10880 S   0.0  0.2  0:00.00 /usr/libexec/wsl-pro-service
204 root         20   0 1715M 12672 10880 S   0.0  0.2  0:00.00 /usr/libexec/wsl-pro-service
Help F2Setup F3Search F4Filter F5Free F6SortBy F7Nice F8Nice + F9Kill F10Quit
```

```
top - 09:42:04 up 1 min, 1 user, load average: 0.08, 0.04, 0.01
Tasks: 23 total, 1 running, 22 sleeping, 0 stopped, 0 zombie
%Cpu(s): 0.0 us, 0.1 sy, 0.0 ni, 99.9 id, 0.0 wa, 0.0 hi, 0.0 si, 0.0 st
MiB Mem : 7636.2 total, 6825.3 free, 473.7 used, 487.2 buff/cache
MiB Swap: 2048.0 total, 2048.0 free, 0.0 used, 7162.5 avail Mem

  PID USER      PR  NI  VIRT   RES   SHR  S  %CPU  %MEM     TIME+ COMMAND
    1 root         20   0 21648 12392 9320 S   0.0   0.2   0:01.06 systemd
    2 root         20   0 3120  1920  1920 S   0.0   0.0   0:00.01 init-systemd(Ub
    6 root         20   0 3120  1792  1792 S   0.0   0.0   0:00.00 init
   42 root        19  -1 50436 18716 17820 S   0.0   0.2   0:00.28 systemd-journal
   93 root         20   0 25284 6400  4992 S   0.0   0.1   0:00.15 systemd-udev
  147 systemd+  20   0 21460 12800 10624 S   0.0   0.2   0:00.10 systemd-resolve
  154 systemd+  20   0 91028 7680  6784 S   0.0   0.1   0:00.06 systemd-timesyn
  163 root         20   0 4236  2560  2432 S   0.0   0.0   0:00.00 cron
  164 message+  20   0 9628  4864  4480 S   0.0   0.1   0:00.05 dbus-daemon
  171 root         20   0 17972  8320  7552 S   0.0   0.1   0:00.08 systemd-logind
  173 root         20   0 1756620 12672 10880 S   0.0   0.2   0:00.09 wsl-pro-service
  176 syslog      20   0 222508 5248  4480 S   0.0   0.1   0:00.06 rsyslogd
  182 root         20   0 3160  1792  1792 S   0.0   0.0   0:00.00 atgetty
  189 root         20   0 3116  1920  1792 S   0.0   0.0   0:00.01 atgetty
  197 root         20   0 107024 22264 13056 S   0.0   0.3   0:00.16 unattended-upgr
  276 root         20   0 3124  768  768 S   0.0   0.0   0:00.00 SessionLeader
  281 root         20   0 3140  1280  1152 S   0.0   0.0   0:00.00 Relay(286)
  286 admin20+  20   0 6072  5248  3584 S   0.0   0.1   0:00.04 bash
  290 root         20   0 6692  4608  3840 S   0.0   0.1   0:00.01 login
  324 admin20+  20   0 20312 11136 9216 S   0.0   0.1   0:00.11 systemd
  325 admin20+  20   0 21160 3520  1792 S   0.0   0.0   0:00.00 (sd-pam)
  349 admin20+  20   0 6072  4992  3456 S   0.0   0.1   0:00.02 bash
  393 admin20+  20   0 9276  5632  3456 R   0.0   0.1   0:00.01 top
```

